

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name Ammonia HR

Product No AC4011-AMP

Pure substance/mixture Mixture

Contains Mercuric (II) Iodide, Sodium Hydroxide

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

Supplier

Fisher Scientific Korea	Thermo Fisher Scientific©
D5,D6, Incheon Airport Logistics	Water and Lab Products
Complex	22 Alpha Road
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Beon-Gil	1-978-232-6000
Jung-Gu, Incheon 400-340	
Tel: +82-2-6196-5500	
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Chem.KR@thermofisher.com info.water@thermo.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
 CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

Made in USA

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Substances/mixtures corrosive to metal

Category 1

AC4011-AMP

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

Health hazards

Acute Dermal Toxicity	Category 3
Acute Inhalation Toxicity - Vapors	Category 4
Skin Corrosion/Irritation	Category 2
Serious Eye Damage/Eye Irritation	Category 2
Reproductive Toxicity	Category 1

Environmental hazards

Chronic aquatic toxicity	Category 3
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Label Elements



Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals
H412 - Harmful to aquatic life with long lasting effects
H311 - Toxic in contact with skin
H315 - Causes skin irritation
H319 - Causes serious eye irritation
H332 - Harmful if inhaled
H360 - May damage fertility or the unborn child

Precautionary Statements

Prevention

P201 - Obtain special instructions before use
P261 - Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray
P264 - Wash face, hands and any exposed skin thoroughly after handling
P271 - Use only outdoors or in a well-ventilated area
P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection
P234 - Keep only in original container
P281 - Use personal protective equipment as required
P273 - Avoid release to the environment

Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P312 - Call a POISON CENTER or doctor/ physician if you feel unwell
P361 - Remove/Take off immediately all contaminated clothing
P363 - Wash contaminated clothing before reuse
P390 - Absorb spillage to prevent material damage
P321 - Specific treatment (see supplemental first aid instructions on this label)
P322 - Specific measures (see supplemental first aid instructions on this label)
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse
P332 + P313 - If skin irritation occurs: Get medical advice/ attention
P362 + P364 - Take off all contaminated clothing and wash it before reuse
P337 + P313 - If eye irritation persists: Get medical advice/ attention
P308 + P313 - IF exposed or concerned: Get medical advice/ attention

Storage

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

P405 - Store locked up

P406 - Store in corrosive resistant stainless steel container with a resistant liner

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

NFPA

Health
3

Flammability
0

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS-No	Index No.	Weight %
Water	No information available	7732-18-5	KE-35400	90 - 95
Sodium Hydroxide	No information available	1310-73-2	KE-31487	1.5 - 2
Sodium Borate	No information available	1303-96-4	KE-03483	1 - 1.5
Mercuric (II) Iodide	No information available	7774-29-0	KE-23126	1 - 1.5
Potassium Iodide	No information available	7681-11-0	KE-29149	0.5 - 1

SECTION 4: FIRST AID MEASURES

First aid measures

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Inhalation

Move to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

Ingestion

Do not induce vomiting. Call a physician or Poison Control Center immediately.

Self-Protection of the First Aider

Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects

None reasonably foreseeable

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

No information available.

Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

Environmental Precautions Should not be released into the environment. See Section 12 for additional ecological information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system.

Methods and Material for Containment and Cleaning Up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling Wear personal protective equipment
Do not get in eyes, on skin, or on clothing
Use only under a chemical fume hood
Do not breathe vapors or spray mist
Do not ingest

Conditions for Safe Storage, Including any Incompatibilities

Storage Keep container tightly closed in a dry and well-ventilated place
Store at room temperature in the original container
Keep away from direct sunlight

Incompatible Products No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

Control Parameters

Component	CAS-No	Korea	ACGIH TLV	OSHA PEL
Water	7732-18-5	Not listed	Not listed	Not listed
Sodium Hydroxide	1310-73-2	Ceiling: 2 mg/m ³	Ceiling: 2 mg/m ³	(Vacated) Ceiling: 2 mg/m ³ TWA: 2 mg/m ³
Sodium Borate	1303-96-4	TWA: 5 mg/m ³	TWA: 2 mg/m ³ STEL: 6 mg/m ³	(Vacated) TWA: 10 mg/m ³
Mercuric (II) Iodide	7774-29-0	Not listed	TWA: 0.025 mg/m ³ TWA: 0.01 ppm Skin	(Vacated) Ceiling: 0.1 mg/m ³
Potassium Iodide	7681-11-0	Not listed	TWA: 0.01 ppm	Not listed

Component	CAS-No	European Union	The United Kingdom	Germany
Water	7732-18-5	Not listed	Not listed	Not listed
Sodium Hydroxide	1310-73-2	Not listed	STEL: 2 mg/m ³ 15 min	Not listed
Sodium Borate	1303-96-4	Not listed	STEL: 15 mg/m ³ 15 min TWA: 5 mg/m ³ 8 hr	Not listed
Mercuric (II) Iodide	7774-29-0	Not listed	TWA: 0.02 mg/m ³ 8 hr	TWA: 0.02 mg/m ³ (8 Stunden). AGW - exposure factor 8 TWA: 0.02 mg/m ³ (8 Stunden). MAK Höhepunkt: 0.16 mg/m ³ Haut
Potassium Iodide	7681-11-0	Not listed	Not listed	Not listed

Biological Exposure Indices

Component	CAS-No	Biological Exposure Indices
Water	7732-18-5	Not listed
Sodium Hydroxide	1310-73-2	Not listed
Sodium Borate	1303-96-4	Not listed
Mercuric (II) Iodide	7774-29-0	Not listed
Potassium Iodide	7681-11-0	Not listed

Appropriate engineering controls

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location

Individual protection measures, such as personal protective equipment

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:
Face-shield.

Skin and Body Protection Wear protective gloves/clothing.

Respiratory Protection None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls Prevent product from entering drains

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State Liquid
Appearance Slightly yellow
Odor Odorless
Odor Threshold No information available

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

pH	13.5
PH Range	11.0 - 14.0

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	100 °C / 212 °F	
Flash Point (High in °C)	N/A	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other Information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No Information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

Inhalation	Not an expected route of exposure.
Ingestion	May be harmful if swallowed.
Eyes	Avoid contact with eyes. Irritating to eyes.
Skin	Avoid contact with skin. May cause irritation. Harmful in contact with skin.

Information on Toxicological Effects

Symptoms No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

(b) skin corrosion/irritation;	Category 2.
Sensitization	No information available
Mutagenic Effects	No information available
Carcinogenicity	No information available.
Reproductive Effects	No information available
STOT - single exposure	No information available
STOT - repeated exposure	No information available
Target Organs	None known.
Aspiration hazard	No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral)	1192 mg/kg
ATEmix (dermal)	332 mg/kg
ATEmix (inhalation-dust/mist)	3.3 mg/L

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

2.5% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Sodium Hydroxide 1310-73-2	-	LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss)	-

Persistence and Degradability

No information available

Bioaccumulation/ Accumulation

No information available

Mobility

.

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

Waste Disposal Methods Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport Not Regulated

IATA

UN-No UN1760
Proper Shipping Name Corrosive liquid, n.o.s. (contains sodium hydroxide and mercuric iodide)
Hazard Class 8
Packing Group II

IMDG/IMO

UN-No UN1760
Proper Shipping Name Corrosive liquid, n.o.s. (contains sodium hydroxide and mercuric iodide)
Hazard Class 8
Packing Group II
Marine Pollutant Not Applicable

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS-No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	-	X
Sodium Hydroxide	1310-73-2	KE-31487	X	215-185-5	X	X	-	X	X	X
Sodium Borate	1303-96-4	KE-03483	X	-	X	X	-	X	X	X
Mercuric (II) Iodide	7774-29-0	KE-23126	X	231-873-8	X	X	-	X	X	X
Potassium Iodide	7681-11-0	KE-29149	X	231-659-4	X	X	-	X	X	X

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	Not applicable	Not applicable	Annex I - Y35
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	X	Annex I - Y29
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Water	7732-18-5	Listed	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Listed	Not applicable	Not applicable
Sodium Borate	1303-96-4	Listed	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS-No	Act on Registration and	Authorised Chemicals	Existing Substances
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SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

		Evaluation of Chemical Substances (K-REACH)		Subject to Registration
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Annex 1 - KE-31487	Not applicable	Listed
Sodium Borate	1303-96-4	Annex 1 - KE-03483	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Annex 1 - KE-23126	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Annex 1 - KE-29149	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	97-1-136 (>5%)	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	97-1-140 (>1%)	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

Component	CAS-No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Listed	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Listed	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	Ceiling: 2 mg/m ³	Not applicable
Sodium Borate	1303-96-4	Not applicable	TWA: 5 mg/m ³	Not applicable

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Formulations containing 0.3% or more by volume (note that aryl and alkyl compounds are excluded from special controlled substances)
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS-No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS-No	Korea	Biological Exposure Indices
Water	7732-18-5	Not listed	Not listed
Sodium Hydroxide	1310-73-2	Ceiling: 2 mg/m ³	Not listed
Sodium Borate	1303-96-4	TWA: 5 mg/m ³	Not listed
Mercuric (II) Iodide	7774-29-0	Not listed	Not listed
Potassium Iodide	7681-11-0	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS-No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Water	7732-18-5	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable
Potassium Iodide	7681-11-0	Not applicable	Not applicable

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Component	CAS-No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Sodium Hydroxide	1310-73-2	Not applicable	1000 lb	Not applicable
Sodium Borate	1303-96-4	Not applicable	Not applicable	Not applicable
Mercuric (II) Iodide	7774-29-0	Not applicable	Not applicable	1.0
Potassium Iodide	7681-11-0	Not applicable	Not applicable	Not applicable

CLP Classification - Regulation (EC) No 1272/2008

Danger.

H311 - Toxic in contact with skin. H332 - Harmful if inhaled. H315 - Causes skin irritation. H319 - Causes serious eye irritation. H360FD - May damage fertility. May damage the unborn child. H412 - Harmful to aquatic life with long lasting effects.

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water. P312 - Call a POISON CENTER or doctor/ physician if you feel unwell. P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing. P337 + P313 - If

SAFETY DATA SHEET

Ammonia HR

Revision Date 05-Sep-2019

eye irritation persists: Get medical advice/ attention. P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection.

SECTION 16: OTHER INFORMATION

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.©
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Reason for revision	Initial Release.

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Disclaimer

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End of Safety Data Sheet